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Physiological basis of the response of harvest index to the fraction of water transpired after anthesis: A simple model to estimate harvest index for determinate species

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ABSTRACT

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The fraction of biomass partitioned to grain is critical to crop yield. Methods to predict this fraction are required for crop modelling. The relationship between harvest index (HI) and the fraction of water transpired after anthesis (θ_E) provides a simple basis for the estimation of HI for determinate species. In this paper, empirical and theoretical aspects of this relationship are examined and a method is developed to estimate HI. The model proposed is:

$$\mathrm{HI}_{pv} = \theta_{E_e} / [1-(a-b\theta_{E_e})] \qquad (1)$$

Subscripts pv and E indicate that HI and θ_E are normalized by production value and vapour pressure deficit, respectively. The model also differentiates between source and sink restrictions to yield. The parameter a quantifies the 'potential' contribution of pre-anthesis assimilates to grain, and $1/(1-a+b)$ the 'potential' harvest index; both are genotype-dependent.

The model accounted for 75% ($P<0.001$) of the variance of HI of wheat plants with HI ranging from 0 to 0.47 and θ_E from 0.05 to 0.47, and 81% ($P<0.001$) of the variance of HI of sunflower plants with HI from 0.13 to 0.46 and θ_E from 0.04 to 0.41.

INTRODUCTION

Grain-yield depends on both biomass production (e.g., Zelitch, 1982; Lambers, 1987) and partitioning of biomass to grain (Snyder and Carlson, 1984). Whereas photosynthesis and transpiration have provided suitable bases for modelling biomass production (Stapper, 1986), no comparably simple, functional methods have been developed to deal with the partitioning of biomass between vegetative and reproductive structures. The only simple ap-

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proach to emerge is the use of empirical partitioning coefficients (e.g. Charles-Edwards, 1982), but these must be determined for each individual condition (Horie, 1977). More mechanistic methods have focused on the partitioning of biomass among vegetative structures (Schulze et al., 1983; France and Thornley, 1984; Stutzel et al., 1988).

The association between harvest index (HI; Donald and Hamblin, 1976) of pot-grown wheat plants and the fraction of water transpired after anthesis (θ_E) found by Passioura (1977) was later confirmed by Richards and Townley-Smith (1987) over a wider range of conditions. Richards and Townley-Smith (1987) found a curvilinear relationship between these variables (cf. Passioura, 1977), and proposed that the curvilinearity resulted from a variable contribution of preanthesis assimilate to yield. Analysis of data published by Merrien et al. (1981), Blanchet and Merrien (1982) and Connor et al. (1985) also showed a significant ($P<0.05$), curvilinear relationship between HI and θ_E for sunflower (Sadras, 1990). This relationship provides a simple and potentially sound method for estimating grain-yield of determinate species in association with either a photosynthesis- or transpiration-based method to estimate biomass production.

This paper presents a physiologically based model of the relationship between HI and θ_E. Because we are working on the agronomy of the sunflower crop, some of its particularities, viz. the relatively high contribution of the pre-anthesis assimilates to grain-filling (Hall et al., 1989) and the high cost of synthesis of the oil and protein components in the seeds (Penning de Vries et al., 1983) are considered.

PHYSIOLOGICAL BASIS OF THE RELATIONSHIP HI VS θ_E

Figure 1 presents an idealized linear relationship between biomass production and transpiration. From this,

$$B_{\mathrm{am}}/B_{\mathrm{em}}=E_{\mathrm{t_{am}}}/E_{\mathrm{t_{em}}} \tag{2}$$

where B and E_{t} are accumulated biomass and accumulated transpiration, respectively, during the periods of anthesis–physiological maturity (am) and emergence–physiological maturity (em). The ratio $E_{\mathrm{t_{am}}}/E_{\mathrm{t_{em}}}$ is the fraction of water transpired after anthesis (θ_{E}).

If the contribution of pre-anthesis assimilate to grain were negligible, and if all the biomass produced after anthesis were partitioned to grain, then the relationship between HI and θ_{E} could be expressed [cf. eq. (2)]

$$\mathrm{HI}=\theta_{\mathrm{E}} \tag{3}$$

Equation (3) is unrealistic, however, because there are two groups of factors that cause deviations from the simple 1 : 1 relationship. The first includes those related to the assumption of linearity between biomass and transpira-

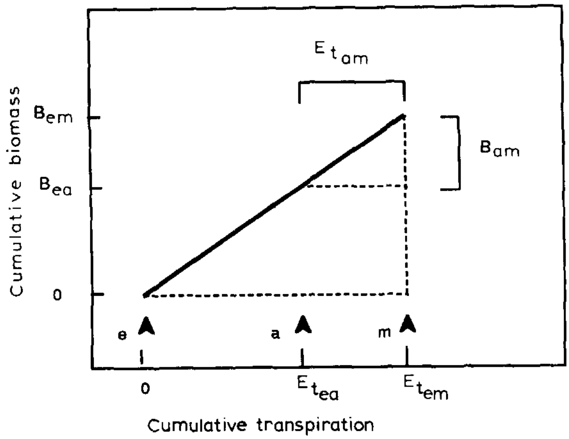

Fig. 1. Idealized representation of the relationship between biomass and transpiration. The arrows indicate emergence (e), anthesis (a) and physiological maturity (m). B and E_t are biomass and transpiration accumulated during different periods, e.g. ea = emergence to anthesis.

tion (Fig. 1). The second concerns the assumptions that all biomass produced after anthesis is partitioned to grain, and that pre-anthesis assimilate do not contribute to grain-yield.

Linearity of the relationship between B *and* E_t

The slope of the curve B vs E_t during the season (Fig. 1) is the transpiration efficiency E_{t_e} of the plant or crop. Factors affecting E_{t_e} will affect the assumption of linearity implicit in the previous equations.

The major factors influencing E_{t_e} are the biochemical pathway of carbon fixation, the vapour pressure deficit (δe), and the biochemical composition of the biomass (Tanner and Sinclair, 1983; Sinclair et al., 1984). For individual species the last two are the relevant factors.

Other sources of intra-seasonal variation in E_{t_e}, however, cannot be discarded and require further evaluation. For instance, the radiation-use efficiency of sunflower crops falls after anthesis (Whitfield et al., 1989) consistent with the loss of nitrogen from the leaves (Hall et al., 1989) and the decay of ribulose 1,5-bisphosphate carboxylase activity (Goswami and Srivastava, 1988). Changes in radiation-use efficiency may or may not influence E_{t_e}, depending upon concurrent changes in transpiration per unit intercepted radiation (Matthews et al., 1988).

Vapour-pressure deficit

Seasonal E_{t_e} is an inverse function of δe (Bierhuizen and Slatyer, 1965;

Tanner and Sinclair, 1983). For crops with starch as a main component in the reserve organs, most of the intra- and inter-seasonal variability in E_{t_e} can be accounted for by normalizing E_t by δe (e.g., Rijtema and Endrodi, 1970; Tanner, 1981; Wilson and Jamieson, 1985) or a related variable such as pan evaporation (e.g. de Wit, 1958; Fischer, 1979).

Biochemical composition of plant tissues

The amount of biomass produced per unit of hexose substrate is the production value (PV) of the biomass (Penning de Vries et al., 1974). Typical PV of vegetative biomass is ca. 0.8 g dry-matter (g^{-1} glucose; e.g. Lafite and Loomis, 1988) compared with about 0.4 for oilseeds such as sunflower (Blanchet and Merrien, 1982) or sesame (McDermit and Loomis, 1981).

Assuming that assimilate produced per unit water transpired (normalized by δe) does not change, then it can be expected that E_{t_e} before and after anthesis will reflect the difference in PV during these two periods. In the case of protein- and oil-rich crops, these differences can substantially reduce E_{t_e} during seed filling.

Variations during the crop cycle in both δe and PV caused significant differences in E_{t_e} between the pre- and post-anthesis periods in sunflower crops (Sadras, 1990).

Implications for the model

The effects of δe and PV (mainly for oil- and protein-rich seed crops) can be considered separately in the pre- (Pre) and post-anthesis (Post) periods by normalizing θ_E by δe, i.e.

$$\theta_{E_e} = (E_{t_{post}}/\delta e_{post}) / [(E_{t_{post}}/\delta e_{post}) + (E_{t_{pre}}/\delta e_{pre})] \qquad (4)$$

and HI by PV, i.e.

$$\mathrm{HI}_{PV} = (Y/\mathrm{PV}_y)/(Y/\mathrm{PV}_y + Y_n/\mathrm{PV}_{y_n}) \qquad (5)$$

where: Y and Y_n are the biomass in seed and non-seed organs, and PV_y and PV_{y_n} the PV of these organs, respectively.

Contribution of pre-anthesis assimilates to yield, and fate of post-anthesis assimilates

When growth after anthesis is restricted by water availability, significant proportions of yield accrue from pre-anthesis assimilates (A_{pre}) (Gallagher et al., 1975; Bidinger et al., 1977; Austin et al., 1980; Richards and Townley-Smith, 1987; Hall et al., 1989). Under these conditions, most of the carbon assimilated after anthesis is partitioned to grain, and yield can be considered source-limited. Consequently,

$$\text{HI} = (B_{\text{am}} + A_{\text{pre}})/B_{\text{em}} \tag{6}$$

The relative contribution of A_{pre} to grain-filling is smaller when water availability after anthesis is unrestricted (Richards and Townley-Smith, 1987; Hall et al., 1989). If water is freely available after anthesis, it has been shown that stems (Kuhbauch and Thome, 1989), roots (O'Toole and Bland, 1987) and other non-seed organs (Hall et al., 1989) are active sinks for carbon until grain-filling is well advanced. Under these conditions, yield of starch-storage sinks such as wheat grains is limited, not by the supply of sucrose, but by the activity of the starch-synthesizing enzymes and the number of plastids in the endosperm (Ho, 1988). Under these conditions, yield could be considered sink-limited, and

$$\text{HI} = [\text{B}_{\text{am}} - (Y_{\text{n}_{\text{am}}})]/B_{\text{em}} \tag{7}$$

where $Y_{\text{n}_{\text{am}}}$ is the biomass produced after anthesis partitioned to non-grain organs.

Equations (6) and (7) correspond to yield limited by source and sink, respectively. These effects can be included in the relationship between HI and θ_E, provided the dependence of A_{pre} and $Y_{\text{n}_{\text{am}}}$ on the pattern of water use can be established. There is little quantitative information on these relationships, so for simplicity we propose that the relative contribution of A_{pre} to yield (Y) decreases linearly with θ_E, i.e.

$$A_{\text{pre}}/Y = c_1 - c_2\theta_E \tag{8}$$

Experimental data by Richards and Townley-Smith (1987) support this relationship for the condition $\theta_E < c_1/c_2$. For the present purpose, we extended the relationship to $\theta_E \geq c_1/c_2$, where A_{pre} is assumed to be 0 (Richards and Townley-Smith, 1987) and the absolute value of $(c_1 - c_2\theta_E)$ is taken to represent the relative amount of post-anthesis biomass partitioned to non-grain organs ($Y_{\text{n}_{\text{am}}}/Y$). This describes an increase of $Y_{\text{n}_{\text{am}}}/Y$ with increasing water availability after anthesis, which is qualitatively consistent with the previous discussion.

A PHYSIOLOGICALLY BASED MODEL OF THE RESPONSE OF HI TO θ_E

From (2), (6) and (8), and normalizing θ_E by δe and HI by PV to maintain linearity between B and E_t, we can write

$$\text{HI}_{\text{PV}} = \theta_{E_e}/[1 - (a - b\theta_{E_e})] \tag{9}$$

where parameters a and b are equivalent to c_1 and c_2 [Eq. (8)], respectively.

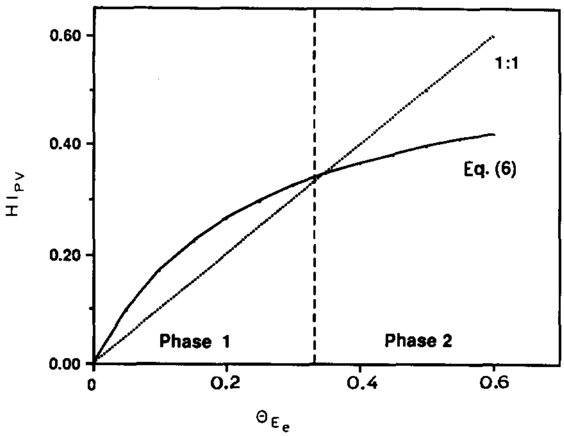

Fig. 2. A model of harvest index as a function of the fraction of water transpired after anthesis. The vertical line separates Phases 1 (source-limited yield) and 2 (sink-limited yield). Values are only indicative. See text for details.

The parameter a is the 'potential' contribution of pre-anthesis assimilates to yield, i.e. the limit of $(a-b\ \theta_{E_e})$ when θ_{E_e} tends to zero [cf. eq. (8)]. The limit of HI_{PV} when θ_{E_e} tends to 1 [Eq. (9)] is $1/(1-a+b)$ and represents the 'potential' (sink-limited) harvest index. The ratio $a{:}b$ has a relevant physiological meaning, as discussed below.

The non-linear model in Eq. (9) is represented in Fig. 2 with the 1:1 relationship [Eqn. (3)] included for comparison. Two phases are defined according to the restriction to yield. In phase 1, yield is limited by source, and in phase 2 it is limited by sink. The division between these phases occurs at $\theta_{E_e}=a/b$ (Fig. 2). According to the model, at this 'equilibrium' point ($HI_{PV}=\theta_{E_e}$) all yield originates from post-anthesis assimilates and all post-anthesis assimilates are partitioned to grain.

In Phase 1, [$\theta_{E_e}<(a/b)$; $HI_{PV}>\theta_{E_e}$] the denominator in Eq. (9) is less than 1 and represents the relative contribution of pre-anthesis assimilates to grain-yield (cf. Richards and Townley-Smith, 1987). In Phase 2, [$\theta_{E_e}>(a/b)$; $HI_{PV}<\theta_{E_e}$], the denominator exceeds 1 by an amount which is taken to indicate the fraction of post-anthesis assimilates allocated to non-grain organs. As explained in the previous section, empirical evidence is required to justify this assumption in the model. The differences between curves in Fig. 2 result from the contribution of pre-anthesis assimilates to grain-yield (Phase 1) and the partition of post-anthesis assimilates to non-grain organs (Phase 2).

EVALUATION OF THE MODEL

Data on HI and θ_E for wheat reported by Richards and Townley-Smith (1987) were used to test the model. These data provided a limited test of the model because, for wheat, PV does not vary substantially with ontogeny, and since the experiments were carried out in a controlled environment, δe was approximately constant throughout the season.

Published data for sunflower demonstrate the non-linearity of the relationship between HI and θ_E (see Introduction) but are not sufficiently extensive to test the model. In order to extend the evaluation to other species with differences in PV between the pre- and post-anthesis periods, an experiment with sunflowers was carried out under field conditions.

Wheat

Method

The data of Richards and Townley-Smith (1987, their tables 2 and 3) were used to test the model. A log function fitted to this subset compared very well with the function presented by those authors for the complete data (Richards and Townley-Smith, 1987, their fig. 2); i.e. the intercepts were -0.11 and -0.10 and the slopes 0.128 and 0.130, respectively. This demonstrates that the data used here form an unbiased sub-sample of the complete data set. As explained before, no variations in PV and δe were considered.

The model [Eq. (9)] was fitted to the data using the FITCURVE procedure of GENSTAT (Lane and Payne, 1987).

Results

The model accounted for 75% of the variance ($P<0.001$) and the mean standardized residual was 0.02 (Fig. 3). Model estimates of parameter b (1.68, SE$=0.209$) and the 'equilibrium point' ($\theta_{E_e}=0.35$) are in agreement with the corresponding values established by Richards and Townley-Smith (1987), of

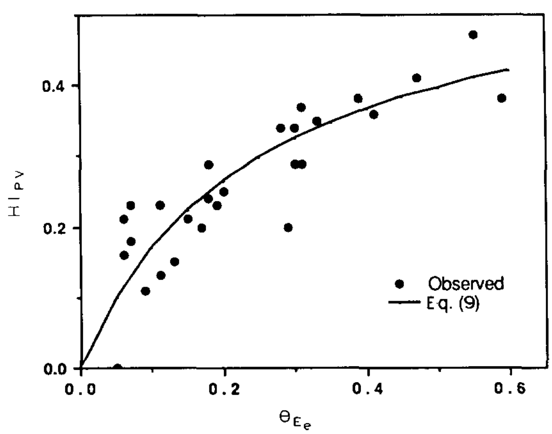

Fig. 3. Model evaluation of the relationship between HI_{PV} and θ_{E_e} for wheat.

1.7–2.1 and 0.34–0.42, respectively. Model estimate of parameter a was smaller (0.58, SE=0.06) than the 0.71–0.72 of Richards and Townley-Smith (1987). However, the method they used over-estimates the contribution of pre-anthesis assimilates to grain-filling (Hall et al., 1989) and may be the cause of their higher a.

Sunflower

Method

The experiment included three hybrids — Cannon (standard height, Cargill); Beauty (semi-dwarf, Cargill); and Dwarf II (dwarf, experimental hybrid, AgSeed). They differ in phenology, HI and contributions of pre-anthesis assimilates to grain (Sadras, 1990; Sadras et al., 1990).

Plants were grown in 55-L containers in the field at the Institute for Irrigation and Salinity Research, Tatura (lat. 36°26′S, long. 145°14′E, alt. 114 m). The soil mixture consisted of top-soil (fine sandy loam) ameliorated with a mixture of sand, vermiculite and peatmoss. Superphosphate and ammonium nitrate were added to provide 20 kg P ha^{-1} and 100 kg N ha^{-1}, respectively. Before planting, the soil in the bins was watered to field capacity with a saturated solution of gypsum.

Three seven-day-old seedlings were planted in each container on 31 December 1988, and thinned to a single plant at full expansion of the second pair of leaves. The containers had plastic lids to shed rain and prevent evaporation. Plants grew through a 0.06-m-diameter hole in the centre of the lid. A PVC pipe 0.05 m diameter sunk ca. 0.30 m into the soil, and extruded 0.05 m over the lid, was used to water the plants. Reference plants, used to estimate maximum transpiration, grew in containers with a 0.07-m layer of scoria at the base to allow drainage. Drainage water was extracted by pump through a second PVC pipe penetrated into the scoria.

The containers were surrounded by a sunflower crop sown 42 days before planting. Additionally, a wind-break was used to reduce the effects of strong winds. The top and north sides of the containers were covered with shadecloth to prevent excessive soil temperature. To prevent insect damage, a systemic insecticide (1 g carbofuran $container^{-1}$) was applied at sowing, anthesis and 20 days after anthesis. To prevent damage by birds, the experimental plot was enclosed in a birdcage made with coarse-mesh nylon net after flowering.

Containers were arranged in three rows (blocks) oriented E–W. Distances between plants were 1.8 m between rows and 0.6 m within row. Three replicates per treatment were included. Cultivars within blocks and treatments within cultivars were assigned randomly.

Nine water treatments were maintained. Maximum transpiration was assessed as the difference between water supply and drainage of plants fully watered each week. The other treatments received 60, 40 and 20% of the maximum transpiration starting two weeks before anthesis (treatments T1–T3) and 100, 80, 60, 40, 20 and 0% starting at last anthesis (T4–T9).

Actual transpiration in the pre- (planting to last anthesis) and post-anthesis (last anthesis to maturity) periods for treatments T1–T9 was assessed as the balance between water supply and change in soil water content during the periods.

Above-ground material was harvested at maturity, and dry-weight of seed and non-seed-organs determined.

Normalized HI was calculated assuming PV of seeds to be 0.42 and PV of non-seed-organs to be 0.80 (Blanchet and Merrien, 1982). Weekly transpiration was corrected by δe derived from dry and wet bulk temperatures recorded daily at 15:00 h on a weather station 440 m from the experimental site.

The model [Eq. (9)] was fitted to the data using the FITCURVE procedure of GENSTAT (Lane and Payne, 1987).

Results

The combination of water regimes and cultivars generated ranges for HI from 0.13 to 0.46, for HI_{PV} from 0.28 to 0.62, and for θ_{E_e} from 0.04 to 0.41. Across cultivars, the model accounted for 81% of the variance ($P<0.001$; Fig. 4) with a mean standardized residual of 0.01.

Table 1 presents the parameters for each cultivar. The parameter a was significantly lower for Beauty than for Cannon and Dwarf II. This is consistent with the differential ability of these cultivars to mobilize pre-anthesis reserves to the grain, as established by Sadras et al. (1990).

The high value of a for sunflower (average across cultivars = 0.86) compared with wheat (0.58) reflects the difference between these species in the potential contribution of pre-anthesis assimilates to yield (Hall et al., 1989).

'Potential' HI_{PV} [i.e. $1/(1-a+b)$] ranked Beauty > Cannon > Dwarf II (Table 1). Beauty and Cannon have similar phenological patterns (data not shown), and the differences were associated with a significantly ($P<0.001$)

TABLE 1

Values (±SE) of the parameters of the model [Eq. (9)], and estimates of potential HI ($(1-a+b)^{-1}$) for three sunflower cultivars

Parameter	Cultivar		
	Cannon	Beauty	Dwarf II
a	0.87 (0.019)	0.81 (0.011)	0.92 (0.014)
b	1.47 (0.084)	1.19 (0.041)	1.86 (0.087)
$(1-a+b)^{-1}$	0.63	0.72	0.52

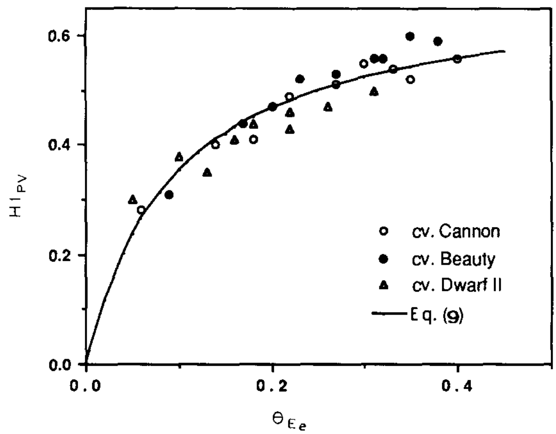

Fig. 4. Model evaluation of the relationship between HI_{PV} and θ_{E_e} for sunflower.

smaller partition of biomass to stems of the semidwarf cultivar. Inverse relationships between HI and plant stature have also been found for cereals (e.g., Johnson et al., 1986; Gent and Kiyimoto, 1989). The smaller 'potential' HI_{PV} of Dwarf II derives from a shorter post-anthesis period, ca. 34% of the total cycle, compared with 41% for Beauty and Cannon. Similar effects of the relative duration of the pre- and post-anthesis periods on HI have been reported (e.g. Done, 1986; Craufurd and Bidinger, 1988).

The intra-specific differences quantified by a and b were not detected when non-normalized HI was fitted to non-normalized θ_E. This was because of the smaller precision of the parameters derived with the non-normalized model. Coefficients of variation (%) derived with the non-normalized model ranged from 11.4 to 24.9 and from 26.6 to 38.7 for a and b, respectively, compared with the normalized model, which gave ranges from 6.4 to 10.2 and from 17.4 to 25.5, respectively.

DISCUSSION

Theory [Eqn. (9)] and experimental data (Figs. 3 and 4) support a simple and consistent relationship between HI and θ_E. The consistency of this relationship requires the consideration of the factors affecting E_{t_e}, primarily δe and PV, which is especially relevant for species with oil- or protein-rich seeds.

A biphasic model that differentiates source and sink restrictions to yield is proposed (Fig. 2). Data by Aggarwal et al. (1986) and Shepherd et al. (1987) indicate that the feedback relationships between source and sink (Neals and Incoll, 1968; Austin, 1989) may impede a clear differentiation between source and sink limitations to yield of droughted crops. Moreover, that yield switches from source- to sink-restricted at a definite 'equilibrium point' (Fig. 2) is probably an oversimplification. However, the differentiation of these phases provided a useful framework to analyze the relationship between HI and θ_E (cf. also Fischer, 1979).

The model fitted well to the data for wheat (Fig. 3) and sunflower plants (Fig. 4). These tests have also shown the consistency of the physiological meaning attributed to the parameters of the model and the sensitivity of these parameters to both inter- and intra-specific variation.

The HI model developed in this paper is proposed as a method to estimate grain-yield of determinate species in conjunction with a photosynthesis- or transpiration-based method to estimate biomass production. The later combination results in a model of yield entirely based on water use, i.e. biomass = f(transpiration); harvest index = f(fraction of transpiration after anthesis); yield = f(biomass, harvest index).

Harvest index of crops under field conditions (Shepherd et al., 1987; Wahbi and Gregory, 1989) seems to be less sensitive to drought than HI of plants in controlled environments (Passioura, 1977; Richards and Townley-Smith, 1987). Thus, caution should be used when extrapolating between these situations.

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